

Mapping Canola, Cereal, and Legume Paddocks Across Australia at 10 m: A Field-Verified, ABS-Validated Dataset

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Draft of September 4, 2026 — PARTIAL Data Descriptor, interim checkpoint (see SUBMISSION_MANIFEST.md)

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Interim checkpoint, not a submission-ready package. This PDF was generated while the national multi-year run (2017–2026, two of nine years complete) is still in progress, while no dataset or code deposit (DOI) yet exists, and while this manuscript has not yet been sent to GRDC for the pre-submission sign-off its data-use agreement requires. Every [PENDING] marker in the source manuscript is preserved below, unfilled. See SUBMISSION_MANIFEST.md for the full readiness statement.

Keywords: crop-type mapping; Sentinel-2; Segment Anything Model; presence detection; accuracy assessment; gradient boosting; Australia

Abstract

We present a national, polygon-level dataset of Canola, Cereal, and Legume paddocks across Australia at 10 m resolution, derived from Sentinel-2 imagery and 2,194 field-verified Grains Research and Development Corporation (GRDC) National Variety Trial records. Paddocks are segmented with the Segment Anything Model and classified into three groups by a gradient-boosted spectral-index model; each polygon carries a predicted class or one of four explicit abstain reasons, a confidence score, and, for Cereal, both raw and Australian Bureau of Statistics (ABS)-calibrated yield estimates. The completed 2024 national release contains 1,346,582 polygons across 99,465 tiles, with a nine-year (2017–2025) 100 km regional companion product demonstrating the pipeline’s multi-year behaviour. Mapped canola composition tracks independent ABS sown-area statistics closely ($r = 0.82$); this dataset is further corroborated by independent spatial comparison against WorldCereal and the National Land Use Map, and discloses and quantifies a known area over-call (1.47–1.59x relative to ABS) concentrated in the Cereal and Legume classes. No other continent-wide, field-verified, paddock-scale crop-species dataset for Australia currently exists.

Keywords

crop-type mapping; Sentinel-2; Segment Anything Model; presence detection; accuracy assessment; gradient boosting; Australia

Background & Summary

Australia’s grains sector — wheat, barley, canola, and pulse/legume crops — is a major export industry with no continent-wide, field-verified, species-resolved crop map publicly available at paddock scale. Official production and area statistics from the Australian Bureau of Statistics (ABS) and the Australian Bureau of Agricultural and Resource Economics and Sciences (ABARES) are available only at coarse administrative-region granularity, with no spatial detail below the Statistical Area Level 2 (SA2) or state level, and the two agencies disagree with each other by a median 6.8% on national sown area — the practical floor against which any finer-grained dataset should be judged. A spatially explicit, species-resolved dataset lets growers, agronomists, and policy analysts see where each crop is actually grown, at paddock scale, rather than inferring it from a regional aggregate.

Existing global crop-mapping products, including WorldCereal (Van Tricht et al., 2023), classify cropland generically or at coarse crop-type resolution and are built around growing-calendar assumptions that do not match Australia’s Southern Hemisphere, canola-and-pulse-rotation grains system; WorldCereal in particular has no Canola or Legume class. The two closest Australian precedents — an in-season Sentinel-2 classifier for six classes in Western Australia (Sharma et al., 2026) and a Sentinel-1/Sentinel-2/MODIS fusion classifier for cereals-versus-canola in the Murray-Darling Basin (Al-Shammari et al., 2024) — are both regional, and both train on either another model’s output or opportunistic harvester-derived labels rather than field-verified ground truth.

This dataset closes that specific gap. It is continent-wide, trains exclusively on GRDC National Variety Trial records — field-recorded sowings, not another model’s predictions or an opportunistic byproduct of harvest operations — and segments paddocks with the Segment Anything Model (SAM; Kirillov et al., 2023) rather than relying on a pre-existing Australian boundary product. Each polygon is classified into one of three broad groups (Canola; Cereal, comprising wheat, barley, and oat; Legume, comprising chickpea, faba bean, field pea, lentil, and lupin) by a gradient-boosted spectral-index model, gated for crop presence, and — for Cereal — attributed with a calibrated yield estimate. Every polygon carries either a predicted class or one of four explicit abstain reasons, a per-polygon confidence score, and, where classified as Cereal, both a raw (National Variety Trial-equivalent) and an ABS-calibrated yield column, so that a downstream user is never left inferring the dataset’s own uncertainty.

The completed 2024 national release contains 1,346,582 segmented polygons across 99,465 tiles (98.9% of the national extent carrying non-trivial agricultural probability for one of the three target crop groups). A nine-year (2017-2025), 100 km regional companion product, covering a mixed-cropping district of the Riverina, New South Wales, demonstrates the identical pipeline’s behaviour across multiple growing seasons, including polygon geometric stability and year-to-year coverage. A national multi-year extension (2017-2023, 2025), applying the identical pipeline nationally across all nine seasons, is in progress at the time of this Data Descriptor and will be released as a dataset update once complete; results describing multi-year behaviour in this Descriptor are accordingly drawn from the regional companion product, not the national extent, and are marked explicitly wherever this applies.

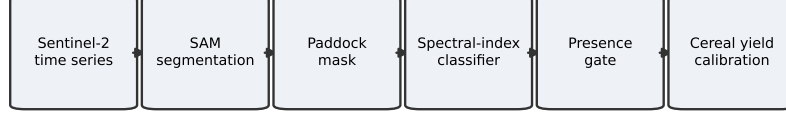
Validated against independent ABS sown-area statistics with no connection to the training pipeline, and independently corroborated by spatial comparison against WorldCereal and the National Land Use Map (NLUM), this dataset provides the first continent-wide, paddock-scale, field-verified crop-species product for the Australian grains sector, together with a quantified, disclosed characterization of its own known limitations — principally an area over-call relative to ABS, concentrated in the Cereal and Legume classes — reported in full in Technical Validation below.

Methods

Fig. 1 gives a pipeline overview alongside a real classified-output panel from the regional companion product, and is a useful orientation before the stage-by-stage description below.

A national, field-verified, paddock-scale crop-species map of Australia

(a) Pipeline overview



(b) Example output: 12km x 12km window, Riverina NSW, 2024

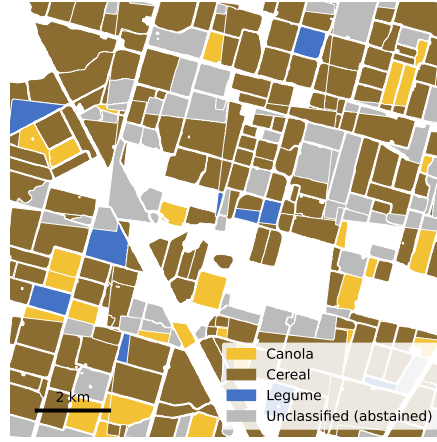


Figure 1: Pipeline overview and example output. Sentinel-2 time series drive both SAM-based paddock segmentation and a 3-class spectral-index classifier; a presence gate determines whether a segmented paddock is classified at all, and a calibrated model attributes Cereal yield to classified paddocks. (a) Six-stage pipeline schematic. (b) Real classified output over a $12\text{km} \times 12\text{km}$ window in the Riverina, NSW (EPSG:3577), 2024 layer of the nine-year regional companion product. 361 polygons — 219 Cereal, 48 Canola, 12 Legume, 82 unclassified (abstained), shown in grey rather than left blank.

Study extent and temporal scope

The national 2024 release covers continent-wide Australia: 99,465 processed tiles, restricted to the extent NLUM v7 indicates carries non-trivial agricultural probability for one of the three target crop groups. A $100\text{ km} \times 100\text{ km}$ block in the Riverina, New South Wales — a major mixed-cropping district — was processed identically across nine seasons (2017-2025) as a regional companion product (Data Records), used throughout this Data Descriptor for multi-year evidence pending the national multi-year extension (below). All spatial processing and reported geometry use EPSG:3577 (GDA94 / Australian Albers Equal Area). The unit of analysis throughout is the individual segmented paddock polygon, not a fixed grid cell — polygon boundaries are themselves an output of the segmentation step described next, not an input constraint.

Data sources

Sentinel-2 surface-reflectance imagery was accessed via Digital Earth Australia’s Analysis Ready Data product (`ga_s2am/bm/cm_ard_3`) through the National Computational Infrastructure’s datacube, at 10 m resolution, nationally, 2017-2025. Ground truth for classifier and yield-model training is drawn from the Grains Research and Development Corporation (GRDC) National Variety Trial (NVT) network: field-verified, point-located records of sown crop per trial site, spanning 2017-2025 nationally, 2,194 records in total. This dataset is covered by a data-use agreement and cannot be released or described at the site level (coordinates, trial identifiers, or yields); all reporting in this Data Descriptor and its accompanying validation is aggregate. Independent validation uses ABS sown-area statistics, aggregated to the SA2 level, which have no connection to the training pipeline; ABARES production statistics provide a secondary cross-check on ABS itself. National Land Use Map (NLUM) v7 250 m per-commodity agricultural probability surfaces (2020-21) were used both to restrict national-scale inference to tiles with non-trivial probability for one of the three target crop groups, and, separately, as an independent post hoc spatial-comparison product (Technical Validation). Paddock boundaries were delineated with the Segment Anything Model (`vit_h` checkpoint, stock parameters; Kirillov et al., 2023) via the SAMGeo/PaddockTS wrapper; re-tuning SAM’s segmentation parameters was investigated as a possible fix for a subset of poorly segmented polygons and ruled out as not cost-effective relative to the presence-gate and area-mask design described below.

Segmentation

SAM was applied per tile at stock parameters; segmented polygons were retained only within a 5-300 ha area range with a compactness ratio ($\text{perimeter} / \sqrt{\text{area}}$) of at most 8 — a free, non-spectral presence signal derived purely from segmentation geometry, which retains 77.9% of known crop paddocks in validation while rejecting 65% of pasture polygons, at a cost of excluding 22% of real crop paddocks that fall outside the size/shape mask.

Feature construction

Fifty-one paddock-median spectral features were computed per retained polygon: spatial medians over 10 m-eroded, cloud-masked paddock pixels per observation date, then temporal medians across dates within 20-day day-of-year bins across the growing season, from three per-pixel indices — NDVI, a canola-specific flowering index following the general approach of Ashourloo et al. (2019), and a chlorophyll-fluorescence-related index.

Labelling

GRDC NVT points were matched to their enclosing (or, where no polygon contained the point, nearest plausible) SAM polygon. Two label-integrity problems were addressed explicitly. First, because NVT trials are agronomic variety comparisons, multiple trials of different crops are frequently co-located within the same physical paddock (69.7% of training trials share a segmented paddock with at least one other trial; 31.8% of those co-located pairs are of different crops), which would otherwise inject direct label noise into paddock-level features. The original nine-species label scheme was collapsed to three broad groups (Canola; Cereal, comprising wheat, barley, and oat; Legume, comprising chickpea, faba bean, field pea, lentil, and lupin) for two separable reasons, both pointing to the same design. First, this resolves 69% of the co-location conflicts by construction, because co-located trials of different species within the same broad group no longer conflict. Second, and independently, a nine-species classifier trained on the same field-verified label pool — using the same features and evaluation protocol as the shipped three-group model — reached only macro F1 0.38, evidently below the label volume this project’s field-verified trial network provides for reliable species-level discrimination. The

three-group scheme reaches macro F1 0.821-0.823 (Technical Validation) on the same underlying population. The remaining ambiguous co-located matches were resolved by hand review of a stratified sample of flagged polygons, which measurably improved classifier performance on canola detection and on temporal transfer (Technical Validation) relative to a size-matched, unreviewed control. Second, a small number of matches to implausibly small polygons (a match falling within a road or laneway sliver rather than the adjacent real paddock) were upgraded to a nearby, at least three times larger polygon within 150 m of the trial point, following a matcher rule shared identically between the training-label-extraction and human-review pipelines.

Classification

Species-group labels are predicted by a `HistGradientBoostingClassifier` operating on the 51 hand-built spectral features described above, evaluated both temporally (training on trials sown 2022 or earlier, testing on 543 fixed trials sown 2023-2024) and spatially (5-fold `GroupKFold` on trial site), to separate temporal from spatial generalization. Fine-tuned Presto embeddings (Tseng et al., 2024) were evaluated as an alternative to hand-built features and were not adopted: at this label volume (2,194 field-verified trials) and class granularity (three broad groups), Presto alone scored macro F1 0.775 against 0.821 for the hand-built feature set, and combining the two gave no measurable improvement over the hand-built features alone (0.824, within the noise of a size-matched control comparison). Three of Presto’s nine expected input channel groups — Sentinel-1 radar, ERA5 climate, and SRTM terrain — were unavailable in this deployment and had to be passed as masked rather than genuinely observed, a plausible explanation for the gap that this dataset’s input availability did not allow closing; the hand-built feature set was retained as the shipped classifier for this reason.

The classifier is trained class-balanced, with geographic location deliberately excluded from the feature set — the model sees only spectral-index time series, not paddock coordinates, so that classification skill cannot be confounded with spatial memorization of where each crop tends to occur. Hyperparameters follow scikit-learn’s `HistGradientBoostingClassifier` defaults with no additional tuning beyond the feature-set and label-scheme decisions described above; no hyperparameter search was run against the ABS validation data, since doing so would compromise ABS’s role as an independent check (Technical Validation).

Presence gating

The shipped presence gate uses NDVI amplitude (90th minus 10th percentile) at a threshold of 0.35. It is fitted presence-only on 3,439 known-sown NVT trials (held-out recall 91.6%) — a larger, presence-only population than the 2,194 field-verified, species-labelled trials used to train the classifier and yield models, since a presence decision only requires knowing a trial was sown, not which of the nine species it was, and more trials meet that weaker requirement. An alternative phenology-shape gate, requiring both a green-up rise and a post-harvest senescence drop rather than amplitude alone, was evaluated on a separate, deployable-variant population (2,177 trials with at least three clear observations) and not adopted. On the 100 km regional companion population, the phenology-shape gate reduced the regional area ratio against ABS from 1.59 to 1.00 (Technical Validation). But per-crop presence recall, on the same 2,177-trial population, shows the shape gate does not fall only on Canola: it reduces Canola recall from 94.5% (amplitude-only) to 78.3% once stacked with the amplitude gate as shipped, and it also reduces Cereal recall (96.0% amplitude-only to 91.4% stacked, −4.6 points) and Legume recall (91.3% to 87.2%, −4.1 points) by a smaller but real margin. Canola’s reduction (16.2 points) is disproportionately larger than Cereal’s or Legume’s, and because Canola is this dataset’s most reliably classified crop and the class whose composition already matches ABS most closely, the amplitude-only gate was retained despite the phenology-shape gate’s superior performance on the area-ratio metric alone. A two-pass variant exempting Canola from the shape gate restores

Canola recall to 94.5% at no measured cost to Cereal or Legume recall but has not been validated against the ABS area ratio and is not part of the shipped pipeline. A third variant, loosening the shape gate’s senescence-drop threshold, was also evaluated:

Gate configuration	Pooled recall	Canola	Cereal	Legume
Amplitude-only (shipped, not stacked)	— (not reported pooled at this population; see per-crop columns)	94.5%	96.0%	91.3%
Phenology-shape, baseline threshold (stacked with amplitude, as shipped comparison)	87.0%	78.3%	91.4%	87.2%
Phenology-shape, loosened senescence (stacked)	89.6%	87.6%	91.6%	87.6%
Phenology-shape, two-pass, Canola exempt (stacked)	91.3%	94.5%	91.4%	87.2%

None of the three phenology-shape variants is the design actually shipped; all three rows are reported here as the evidence base for retaining the simpler amplitude-only gate, not as alternative products. Fig. 6 shows the same comparison graphically, by crop and configuration; the amplitude-only design was selected for the shipped dataset based on this comparison, specifically the Canola-recall cost visible only once recall is examined per crop rather than pooled.

Yield estimation

Cereal yield (pooled wheat, barley, and oat, matching the classifier’s own class boundary) is estimated by a **HistGradientBoostingRegressor** on the same 51-feature contract, calibrated to ABS with a single national multiplicative factor (0.6248) fit on one year of ABS data and checked against two held-out years (+3.1% to +18.1% error). Both raw (NVT-trial-equivalent) and ABS-calibrated yield columns are shipped on every classified Cereal polygon. Sentinel-1 backscatter improves this model too — spatial-transfer R^2 rises by +0.092 when added to the operational satellite-plus-year/state feature set — but is not yet included, for the same national acquisition-cost reason given below for canola.

No canola or legume yield is shipped. For canola specifically, this is a design decision driven by feature availability rather than a fundamentally unworkable target. An optical-only canola yield model’s spatial-transfer coefficient of determination (0.271) does not clear a trivial year-and-state-mean baseline (0.290). The same model with Sentinel-1 backscatter added reaches 0.371, clearly above that baseline — re-tested against the current row-matched population and confirmed, not merely carried forward from an earlier, unrepeatable measurement. Sentinel-1 was also evaluated for the species classifier and was not adopted there: against the current best

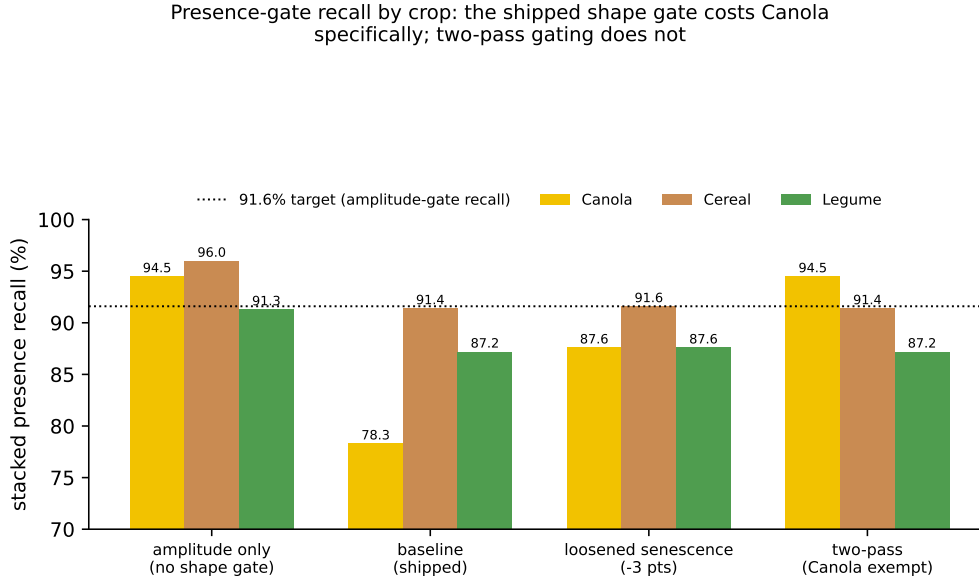


Figure 2: Presence-gate comparison, used to select the shipped gate design. Stacked presence recall (amplitude gate AND presence-shape gate both applied) by crop, across four gate configurations. The amplitude-only design was selected for the shipped dataset based on this comparison, specifically the Canola-recall cost (94.5%→78.3%) visible only once recall is examined per crop rather than pooled; two-pass gating (Canola exempted from the shape gate) restores Canola to 94.5% at no cost to the other classes but is not yet validated against the ABS area ratio.

classifier configuration it regresses macro F1 rather than improving it, the opposite direction from its effect on both yield models. Sentinel-1 is therefore included in neither the shipped classifier nor the shipped yield models at this time; extending its acquisition to national scale, for the yield models specifically, is gated on an ongoing national acquisition-cost assessment and is not yet part of this dataset.

Validation protocol

Composition and area were compared against ABS sown-area statistics across SA2s where the mapped footprint covers at least 50% of the SA2 (133 SA2s nationally); argmax-versus-mean-predicted-probability agreement was used to distinguish genuine classifier behaviour from decision-rule artefacts; a fixed-threshold sensitivity table is reported as a sensitivity check on the presence gate, explicitly not as a fitting procedure against ABS, since fitting to ABS would destroy the only independent validation available. Independently of the ABS comparison, the national map was compared spatially, at the pixel level, against WorldCereal 2021 v100 and NLUM v7 250 m per-commodity probability surfaces, on a stratified random sample of 250,000 of the 1,346,582 national paddock centroids. Full validation results are reported in Technical Validation below.

Multi-year regional companion pipeline

The identical pipeline described above was re-run across nine seasons (2017-2025) over a 100 km × 100 km block in the Riverina, New South Wales, producing the regional companion product used throughout this Data Descriptor for multi-year evidence (polygon geometric stability, year-to-year coverage) pending the national multi-year extension. A national-scale re-run across the same nine seasons is in progress at the time of this Data Descriptor (two of nine seasons

complete) and will be released as a dataset update; no national multi-year numbers are reported in this Data Descriptor ahead of that release.

Data Records

The dataset is released as two products, both in GeoPackage format (EPSG:3577, GDA94 / Australian Albers), described below. [**PENDING — data deposit**]: both products will be assigned a persistent identifier (DOI) at a public repository before submission; this section will be updated with the resolved identifier(s).

National 2024 release

Fig. 2 shows the national tile coverage, the 100 km Riverina regional block, and the SA2 regions used for ABS validation (Technical Validation); Fig. 4 shows the released classes at national extent, rasterised to a 3 km majority-class grid for legibility (individual paddocks are far smaller than one printed pixel at national scale — Fig. 4 is a density visualization for orientation, not the polygon layer itself, which is the actual data record).

The primary product is a single national GeoPackage, 2.8 GB, containing 1,346,582 segmented polygons across 99,465 processed tiles (98.9% of the national extent carrying non-trivial agricultural probability for one of the three target crop groups, per NLUM). Each row is one segmented paddock polygon and carries:

- **pred** — the predicted class (**Canola**, **Cereal**, or **Legume**) where classified, or null where the polygon did not clear the presence gate.
- **abstain_reason** — populated only where **pred** is null; one of four values: **area_below_min** (339,043 polygons), **no_crop_signal** (269,134), **unsegmented_blob** (48,016), or **too_few_observations** (39,623). No polygon is silently dropped: every row in the release carries either a class or a named reason it was not classified.
- **confidence** — the classifier’s predicted-class probability; null where **pred** is null, since a polygon that did not clear the presence gate was never scored by the classifier.
- **ndvi_amp** — the NDVI amplitude (90th minus 10th percentile) used by the presence gate; shipped on every polygon regardless of gate outcome, so a downstream user can apply a stricter threshold than the shipped 0.35 cutoff without re-deriving the underlying signal.
- **yield_tha_raw**, **yield_tha_calibrated** — populated only where **pred** = **Cereal**; raw (NVT-trial-equivalent) and ABS-calibrated Cereal yield in tonnes per hectare, respectively (Methods). No yield fields are populated for Canola or Legume polygons.
- **geometry** — the segmented paddock polygon.

Of the 1,346,582 polygons, 48.4% (34.4% of total mapped area) carry a classification; the remaining 51.6% carry one of the four abstain reasons above. A derived, classified-only subset (650,766 rows, dropping abstained polygons and the **abstain_reason/ndvi_amp** fields) is also provided for users who need only classified paddocks; it should not be used for any claim about abstain rate, total polygon count, or total mapped area, since it silently removes exactly the rows those claims depend on.

Nine-year regional companion (2017-2025, 100 km, Riverina)

A 100 km × 100 km block in the Riverina, New South Wales, processed with the identical pipeline across all nine seasons 2017-2025, is released as nine per-year GeoPackages (one per season) plus a consensus GeoPackage linking each paddock’s identity across years (union-find on inter-year IoU ≥ 0.5, paddock outline taken from the season in which it was largest). This product carries the same per-polygon fields as the national release, plus year and a stability flag

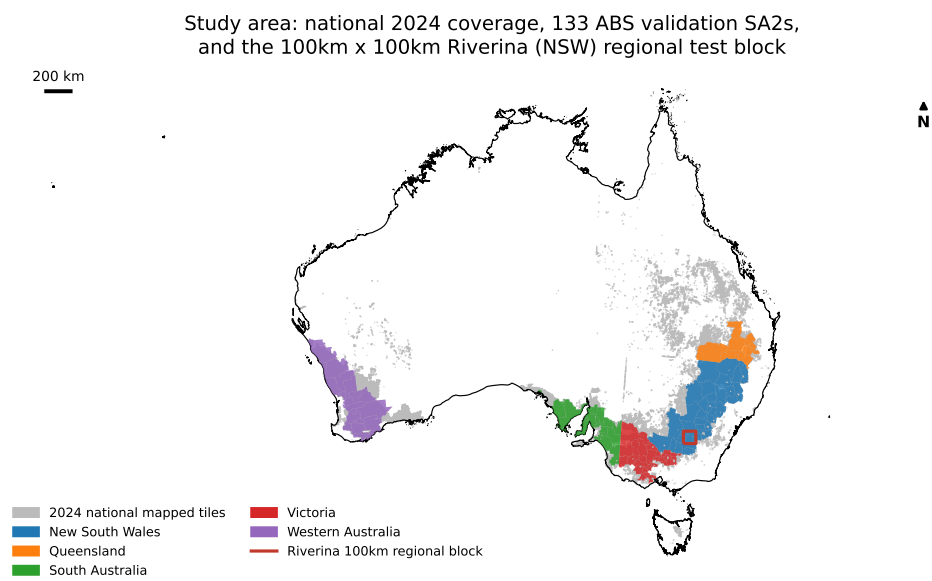


Figure 3: Study area. National extent of 2024 mapped tiles (grey; 99,465 tile centres), the $100\text{km} \times 100\text{km}$ Riverina, NSW block used for the nine-year (2017-2025) regional companion product (red outline), and the SA2s used for ABS composition validation (coloured by state; EPSG:3577). 133 SA2s shown — the $\geq 50\%$ -covered subset re-derived from the published per-SA2 cover table.

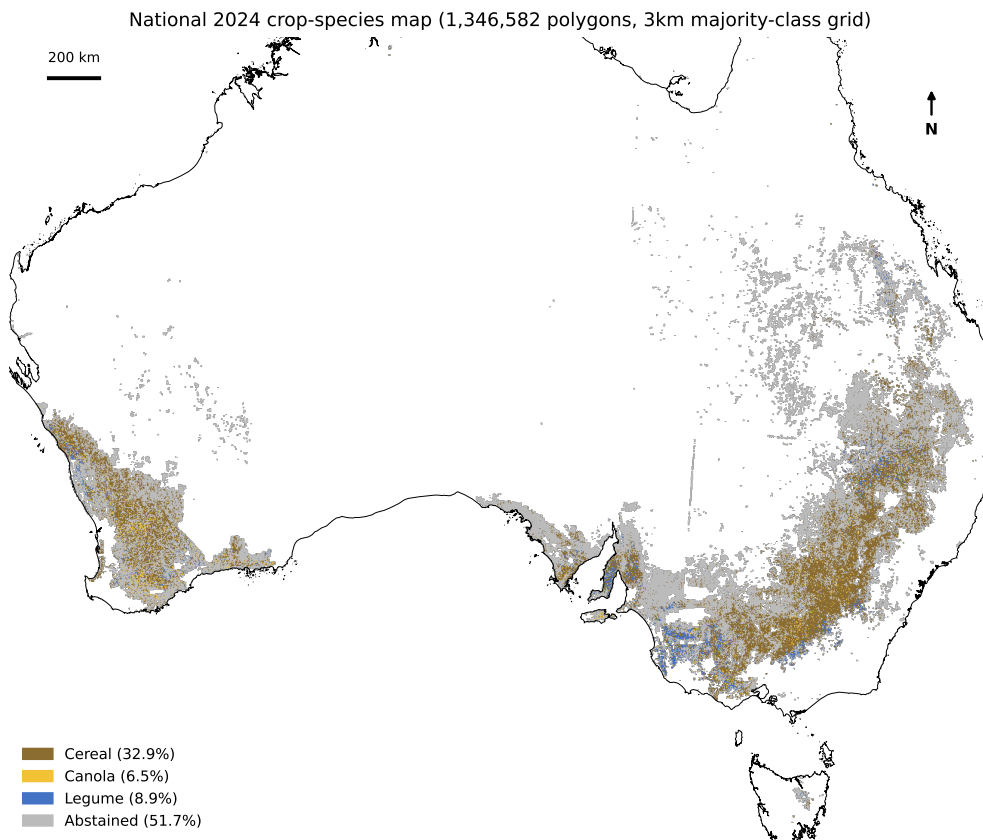


Figure 4: The national 2024 crop-species map. 1,346,582 segmented paddock polygons (EPSG:3577), rasterised to a 3km majority-class grid for legibility at national extent. Classified polygons coloured by predicted class: Cereal 32.9% (443,008), Legume 8.9% (119,578), Canola 6.5% (88,180); unclassified polygons shown in grey (Abstained, 51.7%, 695,816) rather than left blank, carrying one of four abstain reasons. Percentages are by polygon count, not by area.

(found in ≥ 5 of 9 years). It exists specifically to demonstrate multi-year pipeline behaviour — polygon geometric stability, year-to-year coverage — ahead of the national multi-year release described below, and should not be treated as a substitute for national-extent coverage in any of the nine years it spans.

Planned update: national multi-year release (2017-2023, 2025)

A national-scale re-run of the identical pipeline across all nine seasons (2017-2025) is in progress at the time of this Data Descriptor, with two of nine seasons (2023, 2024) complete. Once complete, per-year national GeoPackages with the same schema as the 2024 release above will be added as a dataset update, and this section will be revised to describe them. **[PENDING: national 2017-2025 multi-year run]** — no national multi-year files exist in the current release.

Model artifacts

The classifier (`group3_map.joblib`) and Cereal yield model (`cereal_yield.joblib`) used to produce the national and regional products above are released alongside the data (Code Availability), so that a user can verify or re-derive predictions from the same 51-feature paddock-median input used to generate this release.

Technical Validation

Classifier accuracy

The classifier reaches macro F1 0.821 under temporal transfer (training on trials sown 2022 or earlier, testing on 543 fixed trials sown 2023-2024) and 0.823 under spatial transfer (5-fold `GroupKFold` on trial site), against a 543-trial fixed test set common to both splits. Canola is detected at 89.0% recall at a 5% false-positive rate (average precision 0.944, ROC AUC 0.965).

Split (macro F1)	Class	Precision	Recall	F1	n
Temporal (0.821)	Canola	0.94	0.83	0.88	155
	Cereal	0.89	0.90	0.90	279
	Legume	0.65	0.73	0.69	109
Spatial (0.823)	Canola	0.91	0.86	0.88	155
	Cereal	0.89	0.91	0.90	279
	Legume	0.68	0.70	0.69	109

Legume is the floor in both splits, driven predominantly by Cereal paddocks misclassified as Legume (23 of 279 true Cereal paddocks under temporal transfer, 22 of 279 under spatial transfer) rather than the reverse. Fig. 3 shows both confusion matrices directly.

Composition and area validation against ABS

Validated against ABS sown-area statistics across the 133 SA2s where the mapped footprint covers at least 50% of the SA2, mapped canola composition — its share of total mapped crop area — tracks the ABS canola share closely: $r = 0.82$, median absolute error 5.5 percentage points, below the 6.8% median disagreement between ABS and ABARES that this dataset’s validation treats as the accuracy floor for any map-versus-official-statistic comparison (ABS and ABARES are two official series that themselves disagree by that amount; no validation number tighter than this should be read as more precise than the reference data allows).

Classifier confusion matrices, reviewed arm (n=543, fixed test set both splits)

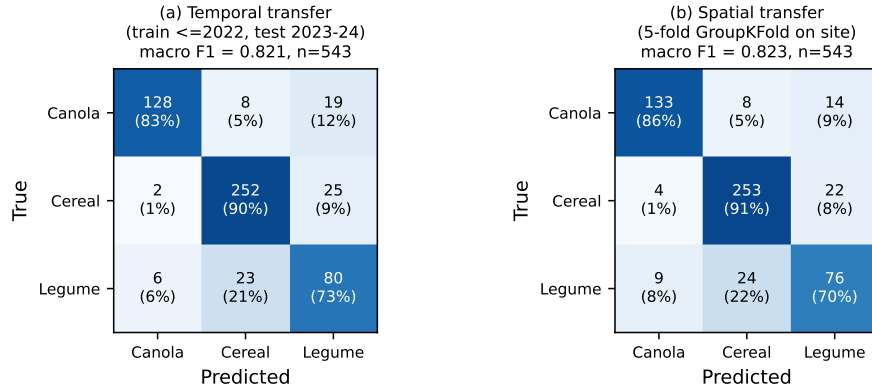


Figure 5: Classifier confusion matrices, reviewed-and-cleaned arm, n=543 fixed test paddocks for both splits. (a) Temporal transfer (train ≤ 2022 , test 2023-24): macro F1 0.821. (b) Spatial transfer (5-fold GroupKFold on site): macro F1 0.823. Legume is the floor in both splits (recall 73% temporal, 70% spatial), driven by Cereal leaking into Legume rather than the reverse.

The 6.8% floor itself is derived directly from ABS and ABARES’s own mutual disagreement, shown here for the three years spanning this dataset’s national release:

Year	Class	ABS area (ha)	ABARES area (ha)	ABS/ABARES ratio
2022	Canola	4,356,211	3,900,000	1.12
2022	Cereal	16,732,981	17,229,002	0.97
2022	Legume	1,849,231	2,154,200	0.86
2023	Canola	3,714,767	3,507,000	1.06
2023	Cereal	14,508,085	16,633,296	0.87
2023	Legume	1,697,392	2,250,103	0.75
2024	Canola	3,671,196	3,438,500	1.07
2024	Cereal	17,627,545	17,733,000	0.99
2024	Legume	3,340,448	3,183,800	1.05

Neither ABS nor ABARES is ground truth; the median disagreement between them (6.8%) is the practical precision floor this dataset’s own validation is measured against throughout.

The dataset’s total mapped area, however, over-calls crop-present land relative to ABS by 1.47x nationally and, independently, 1.59x in the 100 km regional companion — the same mechanism reproduced at two spatial scales.

Class	ABS share	Diluted share	Mapped share	Absorbed (points)
Canola	19.7%	13.4%	13.1%	−0.3
Cereal	67.3%	45.8%	69.0%	+23.2
Legume	8.7%	5.9%	16.7%	+10.8

Diluted is the share ABS would imply if the area over-call were spread proportionally across classes; **absorbed** is the excess mapped area a class actually carries beyond that. The over-call is concentrated almost entirely in Cereal and Legume and essentially absent from Canola: once divided out, canola composition matches almost exactly (13.1% mapped versus a diluted expectation of 13.4%). This pattern — canola composition matching closely, total area over-calling, and the over-call concentrated in Cereal/Legume rather than uniform — indicates the dataset’s dominant source of disagreement with ABS is presence detection (calling too much land crop-present) rather than species misclassification, and this diagnosis reproduces independently at national and 100 km-regional scale with the same class signature both times. Users needing area-accurate rather than composition-accurate output can filter toward a stricter subset using the `dry`, `err`, and `abs` flags.

Fields shipped on every release (Data Records)

tie-break); agreement instead indicates the classifier genuinely, if wrongly, favours Legume at this rate. The mechanism is not established and is disclosed as an open limitation (Usage Notes).

Sensitivity of area to the presence-gate threshold. The shipped gate uses an NDVI-amplitude threshold of 0.35; the same national polygons re-scored at five alternative thresholds show the area/composition trade-off directly, reported here as a sensitivity characterization of the released data, not as a fitting procedure against ABS (fitting the gate to minimize ABS error would make ABS a training target and destroy its value as an independent check):

Threshold	Classified area (ha)	Canola share error (pts)	Legume share error (pts)
0.25	25,509,480	5.9	9.5
0.30	25,509,480	5.9	9.5
0.35 (shipped)	25,509,480	5.9	9.5
0.40	23,528,343	5.8	8.9
0.45	21,374,494	5.5	8.8
0.50	19,015,341	4.8	9.3

Composition error is comparatively insensitive to threshold choice across this range, while classified area is not — consistent with the diagnosis above that the dominant disagreement with ABS is presence detection rather than species discrimination: tightening the presence criterion changes how much land is called crop-present far more than it changes which crop that land is called. Fig. 5 shows the composition-versus-ABS comparison and the area-over-call decomposition together.

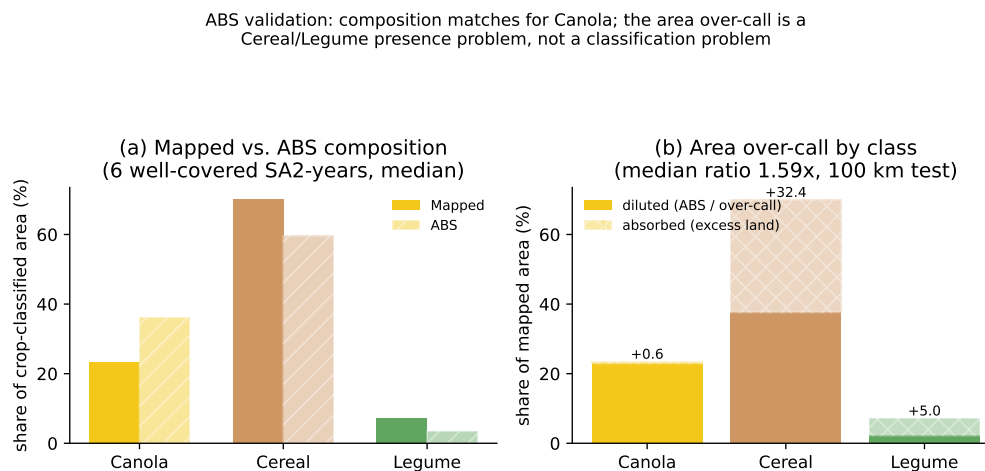


Figure 6: ABS validation: composition and area-inflation decomposition. (a) Mapped vs. ABS crop-share composition (Canola, Cereal, Legume), medians over the well-covered SA2-years. (b) Decomposition of the area over-call by class — ABS share, the diluted share a perfect classifier would report given the measured over-call, and the absorbed excess — showing the over-call sits almost entirely in Cereal and Legume, not Canola.

Independent spatial corroboration: WorldCereal and NLUM

The national map was additionally compared, at the pixel level, against two products with no connection to this dataset’s training or ABS-validation pipeline: WorldCereal 2021 v100 and NLUM v7 250 m per-commodity probability surfaces, on a stratified sample of 250,000 of

the 1,346,582 national paddock centroids. Both comparisons carry a three-year reference-year mismatch (2021 versus this release’s 2024) and are reported with that caveat.

Against WorldCereal’s **temporarycrops** layer, classified-versus-abstained presence agrees at 73.1% overall, symmetrically (73.0% of abstained points also called no-crop by WorldCereal; 73.2% of classified points also called crop). Against WorldCereal’s **wintercereals** layer, Cereal-class agreement is markedly weaker (49.6%, $n=82,411$) — a real disagreement between two independent products, not sampling noise, though neither product is ground truth. NLUM’s per-commodity probability surfaces, covering all three target groups directly, show a consistent enrichment pattern: median NLUM oilseed probability at Canola-classified points is 6.4 times the abstained-point median (the largest ratio in its column), and median legume probability at Legume-classified points is 1.6 times the abstained-point median (likewise the largest in its column). NLUM’s grazing-probability layer independently corroborates the presence-gate mechanism: median grazing probability is 33.8% at abstained points versus 10.8% at classified points, consistent with abstained land preferentially overlapping land NLUM independently rates as probable grazing country.

Cereal yield validation

The Cereal yield model reaches a spatial-transfer RMSE of 1.21 t/ha (30.4% of the 3.97 t/ha median observed yield).

Split	Arm	n	R ²	RMSE (t/ha)
Temporal (train ≤ 2022 , test 2023-24)	Year+state baseline	279	−0.643	2.111
	Satellite features	279	0.471	1.197
	Satellite + year/state	279	0.486	1.181
Spatial (5-fold GroupKFold on site)	Year+state baseline	1,119	0.294	1.473
	Satellite features	1,119	0.526	1.207
	Satellite + year/state	1,119	0.577	1.140

The ABS calibration factor (0.6248, fit on one year of ABS data) checks to within +3.1% to +18.1% of ABS-implied yield on two held-out years. Fig. 8 shows model performance by arm and the calibration check together.

Multi-year regional evidence

Across the nine-year (2017-2025), 100 km regional companion product, classified share ranges from 52.5% (2018) and 59.0% (2017) up to 78.5% (2025); mean NDVI amplitude is correspondingly lowest in 2018 (0.42) and 2017 (0.46) against 0.60-0.62 in 2024-2025, attributable respectively to a documented 2018 drought and Sentinel-2B’s mid-2017 operational start (reduced early-mission revisit frequency and cloud-gap robustness). Polygon geometric stability across the same nine years — independent of any spectral classification — peaks in the 25-100 ha range (the range real paddocks predominantly occupy: median found in 7 of 9 years, median IoU 0.84) and falls at both smaller and larger extremes, an independent, purely geometric corroboration of the area/compactness presence mask (Methods). Fig. 7 shows the year-by-year regional evidence; its national panel is explicitly reserved and left blank pending the national

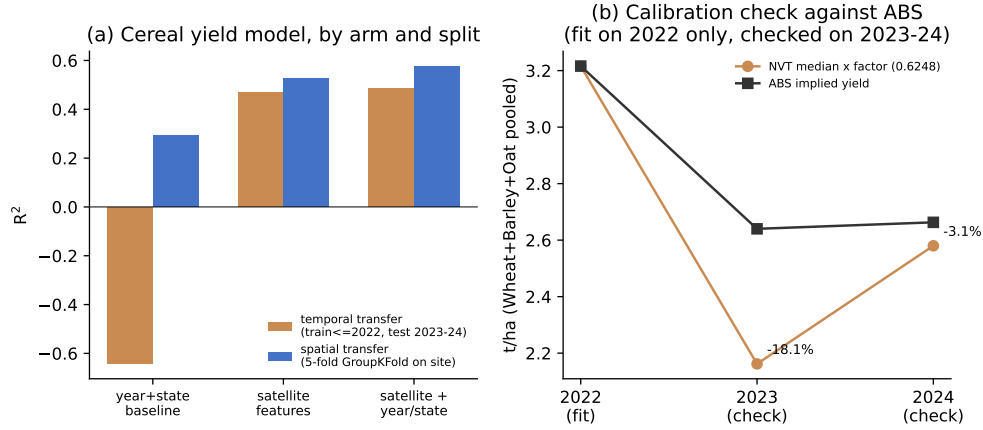


Figure 7: Cereal yield model performance and ABS calibration. Aggregate-only by design — no per-trial predicted-vs-actual scatter is shown, since each point would be a single NVT trial record covered by the GRDC data-use agreement. (a) R^2 by model arm (year+state baseline, satellite features, satellite+year/state) under temporal transfer (train ≤ 2022 , test 2023-24) and spatial transfer (5-fold GroupKFold on site). (b) Calibration check: NVT-median-factor (0.6248) calibrated yield vs. ABS-implied yield, fit on 2022 only and checked on 2023-24 (+3.1% / +18.1% error).

multi-year release (Data Records), not fabricated as a null result. This regional evidence is suggestive of, but does not establish, similar national multi-year behaviour; the national multi-year release is the direct test of that question and is in progress, not yet complete.

[**PENDING — national 2017-2025 multi-year run**]: national multi-year totals, per-year national coverage, and any claim that the single-national-season (2024) validation above generalizes nationally across seasons are not reported here ahead of that release.

Usage Notes

Filtering toward area-accurate rather than composition-accurate output. Because every polygon carries either a predicted class or one of four explicit abstain reasons, plus `ndvi_amp` and classifier `confidence`, a user who needs area-accurate rather than composition-accurate output already has the information required to filter toward a tighter, more conservative view of the dataset without waiting for a future methodological revision. Raising the `ndvi_amp` threshold above the shipped 0.35 cutoff trades completeness for a lower area over-call; Technical Validation’s sensitivity table quantifies this trade-off at several thresholds and should be consulted before choosing a project-specific cutoff. This design was itself informed by an earlier failure mode: an initial design considered building an explicit absence-labelled “non-crop” class from presence-only farmer-recorded grazing records, and was abandoned after finding that 78.5% of a deliberately hard stratum of grazing paddocks passed the same crop-presence test as known crop paddocks — presence-only farmer records could not be reliably inverted into confirmed absence labels. The abstain-reason and confidence fields shipped instead are this dataset’s considered response: rather than force a binary crop/non-crop decision the available negative-label data could not support reliably, every polygon reports its uncertainty explicitly as a first-class field.

The area over-call is the single most consequential characteristic to account for. Users computing absolute crop area from this dataset (rather than composition/share, which is comparatively well validated) should expect a 1.47-1.59x over-call relative to ABS unless they filter using the fields above; the over-call is concentrated in Cereal and Legume and essentially

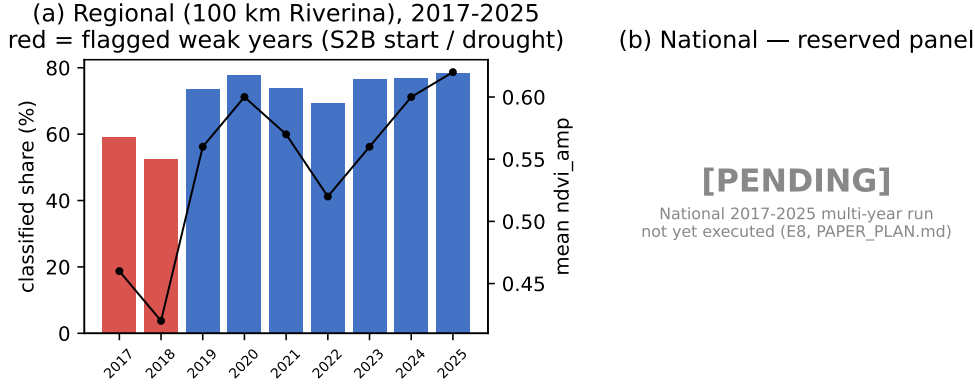


Figure 8: Multi-year regional evidence, with a reserved national panel. (a) Year-by-year classified share and mean `ndvi_amp`, 2017-2025, 100km Riverina box; 2017 and 2018 (red) are flagged as structurally weaker years, attributed to Sentinel-2B’s mid-year operational start and a genuine drought respectively. (b) **[PENDING: national 2017-2025 multi-year run]** — the equivalent national series is reserved but intentionally left blank pending the national multi-year release; do not read panel (b) as a null result.

absent from Canola (Technical Validation), so composition estimates that rely primarily on Canola are the most directly trustworthy use of this dataset as released.

Cereal yield: use both columns, never only the calibrated one. `yield_tha_raw` reflects National Variety Trial management conditions, which generally out-yield commercial paddocks; `yield_tha_calibrated` applies a single national multiplicative factor to correct for this, checked against ABS-implied yield on two held-out years (+3.1% to +18.1% error) but not validated at finer spatial resolution than state level. Users needing paddock-level yield accuracy should treat the calibrated column as indicative, not exact, and consult both columns rather than the calibrated one alone.

No canola or legume yield is shipped, and canola yield’s absence has a specific, addressable cause. Canola yield requires Sentinel-1 backscatter to clear a trivial year-and-state-mean baseline (Methods); Sentinel-1 is not yet part of this dataset’s production pipeline, pending a national acquisition-cost assessment. This is disclosed rather than silently omitted so that a user is not left guessing whether canola yield was attempted and failed outright, or is simply not yet built.

The Legume-class discrepancy (18.4% mapped vs. 8.7% ABS) has no established mechanism and should be treated as an open, disclosed limitation rather than a solved or well-understood one; users relying on Legume-class output specifically should weight it accordingly against Cereal or Canola output.

unsegmented_blob is SAM’s single largest segmentation failure mode (48,016 polygons nationally, Data Records); these polygons carry no class prediction and are not recoverable from this release without a separate segmentation pass.

Sensitive underlying ground truth. The GRDC/NVT trial records used to build and validate this dataset cannot be released at the site level (coordinates, trial identifiers, or yields) under the data-use agreement covering them; this dataset and its accompanying validation reports contain only aggregate statistics derived from those records, never site-level values.

Data Availability

The national 2024 release and the nine-year (2017-2025) 100 km regional companion product described in Data Records will be deposited at a public repository with a persistent identifier

(DOI) before submission. **[PENDING — dataset deposit not yet complete; identifier(s) to be added here once assigned — PAPER_PLAN_SCIDATA.md §24 point 7.]** The underlying GRDC/National Variety Trial ground-truth data used for training and validation — trial coordinates, crop identities, sowing/harvest dates, and yields — cannot be released, in accordance with the data-use agreement covering this dataset; this restriction applies only to the raw trial records and does not restrict release of the resulting map or the aggregate statistics reported in this Data Descriptor. ABS and ABARES reference statistics used for validation are already public and require no release action.

Code Availability

The pipeline code (segmentation, labelling, classification, presence gating, yield calibration, and validation scripts) contains no site-level records and will be released alongside this Data Descriptor. **[PENDING — code repository deposit not yet complete; identifier to be added here.]** The classifier (`group3_map.joblib`) and Cereal yield model (`cereal_yield.joblib`) used to produce the released data are releasable and contain no site-level records. An unused-in-production phenology-shape presence-gate model, evaluated in Methods and not adopted, is retained for reproducibility but is not part of the shipped pipeline.

Author Contributions

[PLACEHOLDER — author contributions not generated by this skill; author list and role assignments to be supplied by the user.]

Competing Interests

[PLACEHOLDER — to be completed by the authors.]

Acknowledgements

[PLACEHOLDER — to be completed by the authors. Should acknowledge the Grains Research and Development Corporation and the National Variety Trials network for the underlying trial data, and the National Computational Infrastructure (NCI) for compute access via Digital Earth Australia.]

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